

SIMULTANEOUS *Go* VIA QUANTUM COLLAPSE

YASHA SAVELYEV

ABSTRACT. We construct a simultaneous combinatorial game *SGo*, which is in a certain mathematical sense a symmetric simultaneization of the classical board game *Go*. *SGo* is in some ways a simpler game than *Go*, as Komi, Ko and suicide rules are removed. On the other hand it has similar dynamics, enabled by certain deterministic “quantum state” reduction, so that state evolution is deterministic. Using the argument of Nash, we show that a symmetric simultaneous combinatorial game has a mixed equilibrium strategy to draw on average, and so *SGo* has one.

1. INTRODUCTION

The game *Go* is an ancient game of local to global geometric principles. It is a sequential game (that is turns alternate between the two players), which has a benefit of having very simple theoretical structure, (mostly) fitting into the subject of combinatorial game theory. But this also has drawbacks, we need an ad hoc rule like Komi to deal with the advantage of Black having the first turn. Moreover, the Ko situation and its cousins like triple Ko, necessitate certain non repetition rules. Although non repetition rules are necessary in end game of many types of games, like chess for example, in *Go* they are necessary even in the early game, which is arguably pathological. Furthermore, this leads to Ko fights, in which the game loses its local character, and which are counterintuitive to many beginners.

At the same time, partly due to its local to global nature *Go* is almost simultaneous. We make this precise by proving that *Go* has a simple (in a precise sense) symmetric simultaneization, see Theorem 2. The construction of this symmetric simultaneization, called *SGo*, is the main thrust of this note.

SGo is a combinatorial symmetric simultaneous game which is playable on a standard *Go* board, although we need a third type of stones that we call red. It is in some ways a simpler game than *Go* as it loses the Komi, Ko and suicide rules, and it has just two rules stone placement and capture (aside from end of game rule). Its game dynamics are very similar to the classical *Go*.

The idea for *SGo* is loosely inspired by quantum mechanics. Rough idea is to branch the game state whenever the moves of the two players do not commute or interfere in classical *Go*. However, by itself this leads to a game with uninteresting dynamics.¹ To get interesting dynamics, we also have “objective reduction of the quantum state”, it is a kind of collapse of the wave function arising deterministically. The term objective reduction is coined by Penrose, see for instance [4].

Using the famous idea of Nash in [3], we show that *SGo* has a mixed equilibrium strategy to draw on average. That said finding this even approximately is likely

¹It is just an example of the “universal symmetrization” of a sequential game, that is discussed in Section 4.3.

only practical on very small boards, due to the unreal size of the strategy space. We can also ask if *SGo* has a strategy to draw not just draw on average.

The classical game *Go* has inspired a wealth of research on combinatorial game theory, see for instance [1], [5]. It also lead to to Conway’s theory of surreal numbers. There is also recent research, see for instance [2] on simultaneous combinatorial games. We hope that *SGo* provides a simple but rich example that can help with further study of such games.

2. GAME RULES

The game *SGo* is initialized with an empty standard *Go* board, with horizontal/vertical lines whose intersections we just call intersections. There are two players that we call Black and White. As in *Go*, players Black and White move on a given turn by placing a black respectively white stone on an empty intersection on the board, but in *SGo* they do this on the same turn without knowledge of the other’s move on this turn. We allow pass moves as in *Go*. Simultaneous action, requires modifying the placement and capture rules of *Go*, while removing the Komi rule, Ko, and suicide rules. We also need to introduce a third type of stones called red, which may be thought of as simultaneously white and black.

Remark 1. *Note that our rules are not totally ad hoc, the construction of SGo is such that*

- (1) *It forms a simple symmetrization of Go, in the precise sense of Theorem 2. In particular, the “causal reduction rule”, that is to be discussed, is by necessity of satisfying this condition.*
- (2) *It is playable on a standard Go board, which in particular requires us to represent simultaneously moves to the same position, and we choose to do this by introducing red stones. In this case there is no choice, any two ways to represent such simultaneous action are clearly equivalent.*
- (3) *It has similar game dynamics as the standard Go, that is with nearly optimal play the positions tend to become very sharp, that is have high sensitivity. This is far from the case for the universal symmetrization, as considered in Section 4.3. In that case most games are easily drawn, which is not hard to see. To achieve these dynamics we need a way to aggressively prune branched states resulting from simultaneous action. We do this via our capture resolution rules.*

Of course different choices might be been made, that still achieve the above goals, however the interesting fact is that it can be done at all. For example for chess, this looks untenable, see discussion around Question 2.

2.1. Some definitions. Given a board state S , that is any (at the moment no legality restrictions) collection of white, black, red stones on the board, let $G(S)$ denote the graph with vertices stones, and an edge between vertices whenever the corresponding stones are horizontally or vertically adjacent. (Meaning they are adjacent and on the same horizontal/vertical line.) From now on adjacency always means this type of adjacency. In what follows the terms stones and vertices may be used synonymously.

A **unicolor component** C is a connected component of the full (a.k.a. induced) subgraph of $G(S)$ whose vertices are either all the black, or all the white stones. (Full subgraph means that all the edges of the original graph, between any pair

of vertices of the subgraph, are included.) In this case we call C **black**, respectively **white**. We may less often refer to **red components**, analogously defined as connected components of red stones.

As in classical *Go*, a connected component C **has no liberties** if it is not adjacent to an empty intersection (this is short for C does not contain a stone adjacent to an empty intersection). We will also call C a **trapped component**.

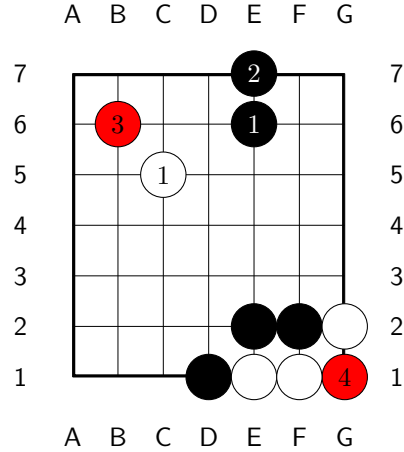
We define stones **adjacent** to C , as those stones whose corresponding vertices are connected by an edge to C in $G(S)$.

Given a unicolor component C , a white/black stone on the board is called a **trapping** if it is adjacent to a trapped unicolor component of opposite color.

2.2. Stone placement resolution rule of *SGo*. If the players move to the same intersection i on the board we place a stone there whose color resolves in the following decreasing order of priority:

- (1) If either a white or black stone placed at i is trapping, then a red stone is placed at i . We call this a **trapping move**.
- (2) If the intersection i is adjacent to a black stone, and neither a white stone nor red stone, a black stone is placed. If the intersection is adjacent to a white stone and neither a black stone nor red stone, a white stone is placed.
- (3) Otherwise, a red stone is placed.

To the right is an example of placement resolution. On move two both players move to E7, the resulting stone resolves to black by our rules. Note that the stone placed on move 4 does not resolve to white, as it is trapping.



2.3. Capture rule of *SGo*. After both White and Black move on current turn numbered n , using the stone placement resolution rule above we get some board state S .

Definition 1. We say that a black/white/red component C is **trapped on turn n** if C is trapped and some stone adjacent to C , or in C was placed on turn n . A stone is called **trapped** on turn n , if it is an element of C as above. We say that a red stone is trapped by white respectively black on turn n , if replacing it by a black respectively white stone it is trapped on turn n .

A white component C is resolved as **captured** on this turn if the following is satisfied:

- (1) C is trapped on this turn.
- (2) If C is adjacent to a red stone that is adjacent to another trapped component then that component is white.
- (3) One of the following holds:
 - (a) No black or red stone adjacent to C is trapped by white on this turn.
 - (b) (Suicidal capture) No stone in C is created on this turn. If a stone in C is adjacent to a stone trapped on this turn, then the latter stone is created on this turn and is not red.

In the case C is captured.

- The stones of C are removed as prisoners of Black as in *Go*.
- Any red stones adjacent to C resolve as black. (Concretely, they are replaced by black stones.)

If C is a black component the process is naturally mirrored. The above capture rule is invoked recursively until no captured on this turn n , unicolor components remain.

2.4. Causal reduction. At the end of the capture procedure, we further simplify the position as follows. If any trapped red component C is adjacent only to white stones, and non of these white stones are trapped then all stones of C are replaced by white stones. This is naturally mirrored in the case C is adjacent only to black stones. The reason we call this causal reduction is outlined in Section 3.2.1.

Remark 2. *This additional rule is logically necessary if we want SGo to be a symmetrization of Go, see Section 4.2. More specifically, condition 3 of Definition 7 will not be satisfied, as the image of natural map ρ in the proof of Theorem 2, would contain an illegal Go history.*

2.5. Stage one reduction. The above capture procedure followed by causal reduction is called **stage one reduction** and the resulting board state will be denoted by $R_n(S)$.

Lemma 3. *The stage one reduction is well defined, that is independent of the order in which components are captured.*

Proof. Suppose we have two unicolor components C_0, C_1 captured on the given turn. By assumption if a red stone is adjacent to both C_0 and C_1 then C_0 and C_1 have the same color. But in that case it clearly does not matter in which order they are captured.

Thus suppose that no red stone is adjacent to both C_0 and C_1 . If C_0 and C_1 have the same color they cannot be adjacent, that is there is no pair of stones, one in C_0 the other in C_1 that are adjacent. We now show that if they have opposite color then they are also not adjacent. Given this, if we apply capture rule to the group C_0 and then to the group C_1 , this results in the same board state as first applying the capture rule to the group C_1 and then the group C_0 .

We have the following possibilities. If C_0, C_1 both satisfy condition 3a, then they are trivially not adjacent. The same happens if C_0 satisfies the former condition and C_1 satisfies condition 3b. Suppose lastly that C_0, C_1 both satisfy condition 3b. And suppose by contradiction that they are adjacent. Then C_1 is adjacent to a stone in C_0 , which by assumptions is trapped on this turn. Condition 3b applied to C_1 implies that the latter stone is created on this turn. But by assumptions no

stone of C_0 is created on this turn. So we have a contradiction, and so C_0, C_1 are not adjacent.

To prove the general case proceed by induction. Let $S(n)$ be the sentence: for a collection $\{C_0, \dots, C_n\}$ of captured unicolor components, the result of stage one reduction does not depend on the order of C_i . $S(1)$ was shown to hold. We show $S(n) \implies S(n+1)$. Let $\{C_0, \dots, C_{n+1}\}$ be a given ordered collection of captured unicolor components, and let ρ be a permutation of the set $\{0, \dots, n+1\}$. By the argument above and using a sequence of adjacent transpositions, the stage 1 reduction applied to $\{C_{\rho(0)}, \dots, C_{\rho(n+1)}\}$ (respecting the given order) is the stage 1 reduction applied to $\{C_0, C_{\rho'(1)}, \dots, C_{\rho'(n+1)}\}$ for some permutation ρ' fixing 0. Using the induction hypothesis on $\{C_{\rho'(1)}, \dots, C_{\rho'(n+1)}\}$ we get that the latter stage 1 reduction is the stage 1 reduction applied to the order collection $\{C_0, \dots, C_{n+1}\}$. $\square$

If there are no unicolor components trapped on the turn $n-1$ or lower, the turn is resolved. Otherwise, we recursively define higher stage reduction as follows.

Define stage k reduction by the condition that the board state after stage k reduction denoted by $R^k(S)$ satisfies:

$$\begin{aligned} R^1(S) &= R_n(S). \\ R^k(S) &= R_{n-k+1}(R^{k-1}(S)). \end{aligned}$$

If there are no unicolor components trapped on the turn $n-k$ or lower, the capture stage is resolved after stage k reduction.

Remark 4. *In practice the resolution of most turns is easy to work out mentally, we will give some examples of such “nested entanglement states” further ahead. Of course having a computer work out turn resolution automatically is very helpful.*

2.6. End of the game. An *SGo* game ends if one of the following holds:

- The players agree to end the game, in this case they may also agree to remove some stones on the board as prisoners for either player. (As is common in standard play of *Go*.) This is meant to be the main way of ending the game.
- The board position at the end of the turn is identical to the board position at the end of the previous turn. For example if one player passes and the other does a single stone suicide on the same turn. (This is meant to insure that a winning player can actually force a win when the opponent is stalling.)

We then decide the winner using Chinese *Go* style rules. So the winner is decided by the integer sum of the prisoner count and area count. We now explain the latter.

We say that an empty intersection i on the board **reaches black respectively white** if there is a connected component of empty intersections (as usual meaning a connected component of the graph with vertices empty intersections on the board and edges adjacencies) that contains i and contains an intersection adjacent to a black stone respectively white stone.

We say i is **surrounded by black** if:

- i reaches black.
- i does not reach white. (To emphasize it could reach a red stone.)

We define surrounded by white analogously. Then in the case of player Black the area count is the sum of:

- (1) The number of black stones on the board.
- (2) The number of red stones adjacent to an empty intersection surrounded by black.
- (3) The number of empty intersections surrounded by black.

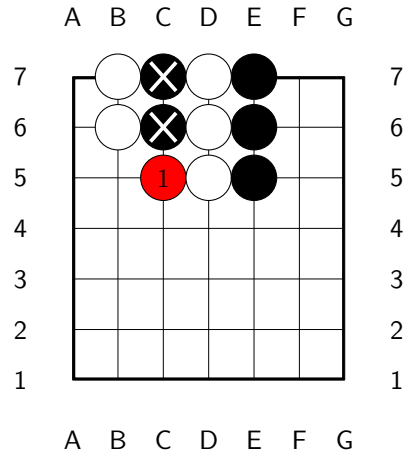
See Section 3.4 for an example.

A draw is now possible although compared to chess it must be much less likely, (on a typical 13×13 or 19×19 go board).

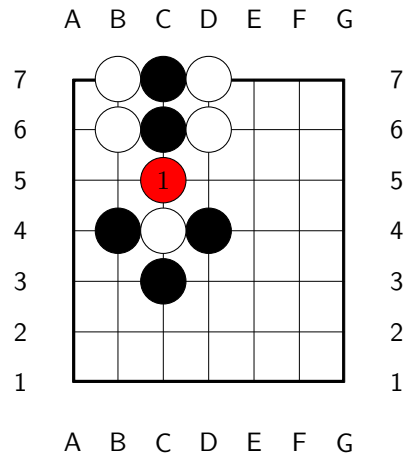
3. EXAMPLES

3.1. Capture and non-capture moves. Here are some examples and non-examples of capture.

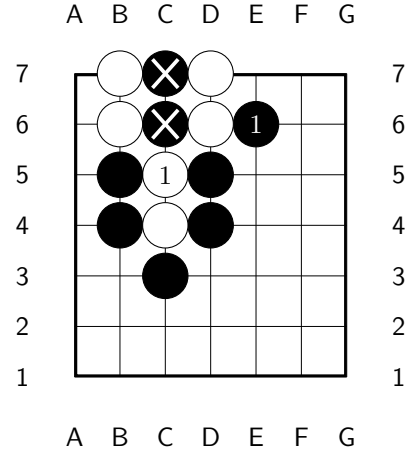
On move 1 both Black and White move to C5. As this is a capture move, C5 will resolve to white at the end of the turn.



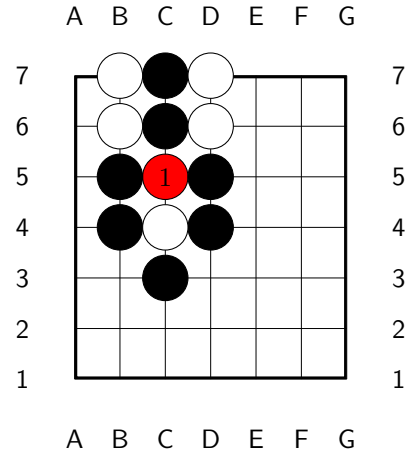
Not a capture move, as the created red stone is adjacent to an opposite color trapped component. This should make intuitive sense since any capture would be ambiguous from classical *Go* perspective.



White's move is a capture move, as Condition 3b is satisfied. This is consistent with classical *Go* behavior that suicidal captures resolve to capture.

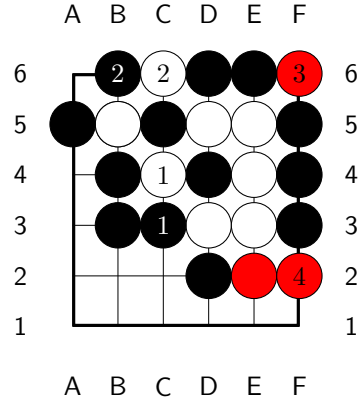


Not a capture move, now Condition 3b is not satisfied, since the stone placed on this turn is red.

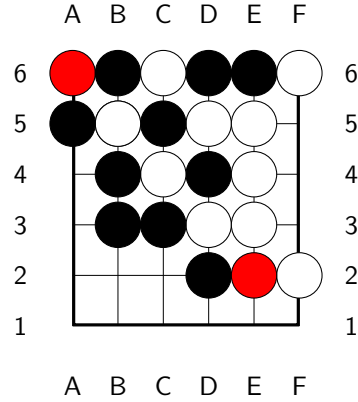


3.2. More complex examples. In the first example, White need only play with a pure strategy (and still be ostensibly winning). In more marginal, complex positions, the players may need mixed strategies.

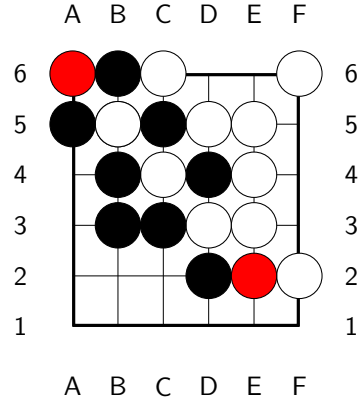
In the position on the right, on turn 1, White attempts capture at C4, while Black attempts capture at C3. Nothing is captured on turn 1 by our rules, and after turn 2 the position becomes an example of a “nested entangled state”. A possible continuation is indicated. Note that F6 is not a capture for White by our rules, similarly to the last example of the previous section. But F2 is a capture move for White since condition 3a is satisfied.



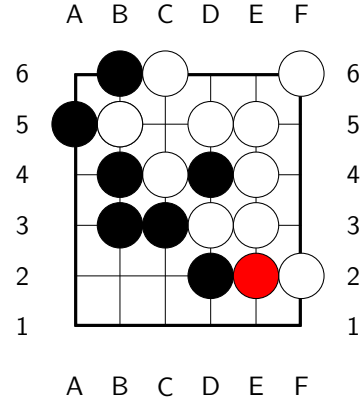
At the end of turn 4, stage one resolution is depicted on the right.



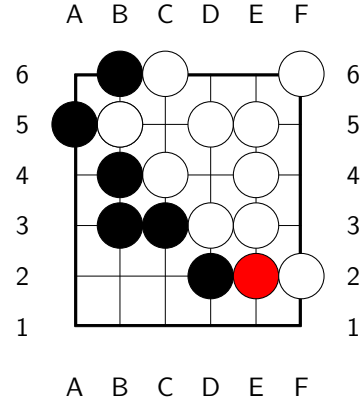
Stage two resolution.



Stage three resolution.

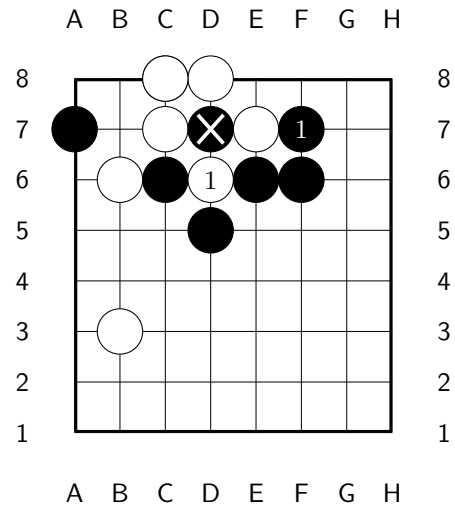


Stage four and final resolution of turn 4.

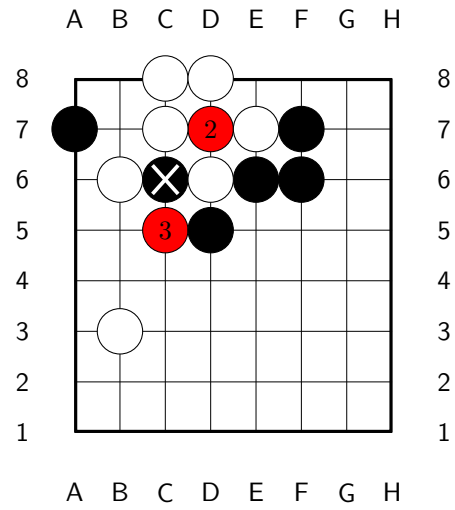


3.2.1. *Ko situation.* Below is a Ko type situation. Recall, that we don't have a Ko rule, but it is now unnecessary, since the board state cannot immediately loop.

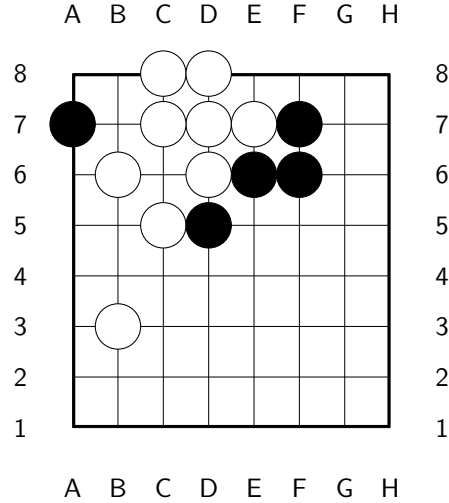
White's D6 is a capture move.



Here is a possible continuation.

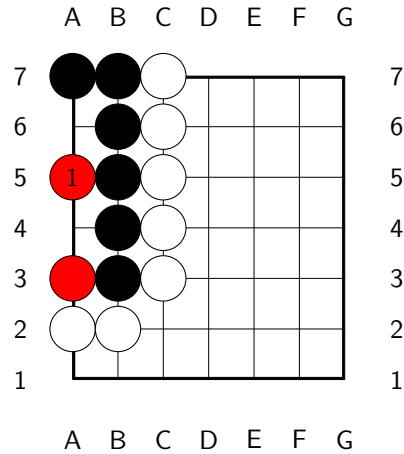


At the end of turn 3 we get the following position. The red stone at D7 is converted to white by the causal reduction rule. To remark, this should make sense from our perspective of *SGo* as a “quantum analogue” of *Go*. A red stone at D7 is meant to be both black and white, but if it was a black stone we would have an illegal *Go* position, so it must actually be white. (This is the meaning behind the term causal reduction.)

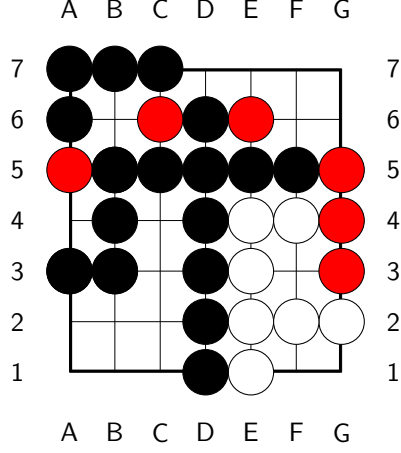


3.3. Life and death. Life and death is positionally very similar to *Go*, here is an example.

The black group is alive, that is impossible to kill.



3.4. End game position. In the diagram below, both players have passed on the current turn and the game has ended. Assuming there are no prisoners, White has $7 + 3 + 1$ points and Black has $7 * 7 - 7 - 3 - 2$ points.



4. SGO AS A SYMMETRIC SIMULTANEOUS COMBINATORIAL GAME

We will use the formalism of automata, as it is very flexible. The goal is to introduce symmetric simultaneous combinatorial games or *ssc* games.

Definition 2. A combinatorial simultaneous game G with 2 players P_0, P_1 or Black, White, consists of:

- A deterministic automaton G , whose set of states is a set $S(G)$ called the set of game states of G .
- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \sqcup M_1(G)$: interpreted as possible moves of Black and White.
- Games states decompose as

$$S(G) = B(G) \sqcup B(G) \times M_0(G) \sqcup B(G) \times M_1(G),$$

where $B(G)$ is some set (for example it might be the set of board states).

- Nonempty subsets $W_0(G), W_1(G), D(G) \subset B(G)$ of final (accepting) states with empty pairwise intersection, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in B(G)$, understood as the state in which the game is initialized.
- A partial game evolution function:

$$E = E_G : S(G) \times M(G) \rightarrow S(G),$$

which has the properties:

- (1) $E(s, m)$ is undefined for s in one of the accepting states.
- (2) If $s \in B(G) \subset S(G)$, then $E(s, m) = (s, m)$ for all $m \in M(G)$ s.t. $E(s, m)$ is defined.
- (3) If $(s, m) \in B(G) \times M(G)$, then $E((s, m), m') \in B(G)$ if defined, and is undefined if the grading of m is the grading of m' .
- (4) Suppose that $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, $s \in B(G)$, $E(s, m_0)$ is defined and $E(s, m_1)$ is defined then $E(s, m_0, m_1)$ is defined.

- (5) For s, m_0, m_1 as in the previous property, if $E(s, m_0, m_1)$ is defined then so is $E(s, m_1, m_0)$ and

$$E(s, m_0, m_1) = E(s, m_1, m_0).$$

We recursively define:

$$E(s, m_1, \dots, m_k) = E(E(s, m_1, \dots, m_{k-1}), m_k).$$

We say that G is **finite** if $S(G), M(G)$ are finite, and all game sequences have bounded length, where a game sequence is a sequence $\{m_i\}_{i=1}^{i=n}$ of moves s.t.

$$\forall 1 \leq k \leq n \ E(q_0, m_1, \dots, m_k) \text{ is defined.}$$

From now we just say simultaneous game for G as above.

The above notion is more interesting when G also satisfies a certain symmetry between the two players.

Definition 3. We say that a simultaneous game G is **symmetric** if the following conditions are satisfied.

- (1) There is a bijection $\mathcal{S} : M(G) \rightarrow M(G)$ taking $M_0(G)$ onto $M_1(G)$.
- (2) There is a bijection $R : B(G) \rightarrow B(G)$ s.t. for all $s \in B(G)$, all $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, and for $\mathcal{S}$ as above we have:
 - $E(R(s), \mathcal{S}(m_0), \mathcal{S}(m_1)) = R(E(s, m_0, m_1))$, whenever both sides are defined. Furthermore, if one side is defined then so is the other.
 - R also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.

Lemma 5. $SG\circ$ is an ssc game, that is a symmetric simultaneous combinatorial game.

Proof. Set $G = SG\circ$, with latter described in Section 2. The set $M_0(G)$ corresponds to possible moves of Black and $M_1(G)$ the moves of White, as in the game description.

The state space is:

$$S(G) = B(G) \sqcup B(G) \times M_0(G) \sqcup B(G) \times M_1(G),$$

where an element of $B(G)$ is a legal board position and its entire legal game history. By legal, we mean the history is obeying the rules of Section 2.

The evolution map E is then obviously defined. For example if $s \in B(G)$ and $m_0 \in M_0(G)$, $m_1 \in M_1(G)$ are legal moves then $E((s, m_0), m_1) \in B(G)$ is defined to be the resolution of the board position after applying the rules of Section 2, once Black's move m_0 , and White's move m_1 are reflected on the board.

To see that G is symmetric, let $R : B(G) \rightarrow B(G)$ be the map that interchanges the black and white colors of the stones of the board states, leaving red stones invariant. And let $\mathcal{S} : M(G) \rightarrow M(G)$ interchange black and white moves, i.e. we have a natural isomorphism $M_0(G) \rightarrow M_1(G)$, which then induces $\mathcal{S}$. Then clearly the conditions of Definition 3 will be satisfied. $\square$

4.1. Solvability. A theorem of Zermelo [6] says that for every suitably finite sequential game either Black or White has a winning strategy or there is a draw strategy. Obviously $SG\circ$ cannot have a winning strategy, furthermore unless we

put an upper bound on the length of a game, or on the number of position repetitions an *SGo* game length can be infinite.² So we will assume there is such a bound, which will make *SGo* a finite game.

Question 1. *Does SGo have a strategy to draw?*

Of course we have simultaneous games with no strategy to draw, the classical example is rock-paper-scissors. But the latter has a strategy to draw on average. The latter being a very trivial case of the existence of the Von-Neumann, Nash equilibrium for “matrix payoff games”. *SGo* is not a matrix payoff game, but Nash’s proof of the equilibrium theorem readily extends to this case. In general:

Theorem 1. *Assigning any payoff for a win and draw, a finite simultaneous game G in the sense above has a Nash equilibrium.*

Proof. For $s \in B(G)$ set

$$M_i(s) = \{m \in M_i(G) \mid E(s, m) \text{ is defined}\}.$$

The space of strategies of player P_i is then:

$$\mathcal{P}_i = \prod_{s \in B(G)} \Delta(M_i(s)),$$

where $\Delta(M_i(s))$ is the space of probability distributions over the finite set $M_i(s)$. Since $B(G)$ is also finite, the joint strategy space $\mathcal{P}_0 \times \mathcal{P}_1$ is a compact convex subspace of $\mathbb{R}^N$, for some very large N .

The expected payoff function on $\mathcal{P}_0 \times \mathcal{P}_1$ is easily seen to be continuous, due to finiteness of G .³ Using this, we get that the multi valued function ϕ , assigning to a joint strategy its set of countering strategies, has a closed graph. Furthermore, the image by ϕ of any point is non-empty and convex.

Then as argued by Nash [3] using Kakutani’s fixed point theorem we get existence of an equilibrium point. $\square$

Corollary 1. *A finite ssc game G has a strategy to draw on average, and in particular SGo has such a strategy.*

Proof. Assign 1 for a win, -1 for a loss and 0 to a draw. Then it is enough to show that for any Nash equilibrium strategy the expected payoff is 0 for both players.

Suppose otherwise, and suppose without loss of generality that

$$(\mathcal{D}_0, \mathcal{D}_1) \in \mathcal{P}_0 \times \mathcal{P}_1$$

is an equilibrium joint strategy where Black has a positive expected payoff.

Let R, S be the symmetry operators as in the proof of Lemma 5. Naturally applying these operators to the joint strategy $(\mathcal{D}_0, \mathcal{D}_1)$ we get that there is an equilibrium joint strategy $(\mathcal{D}'_0, \mathcal{D}'_1)$ where White has positive expected payoff. But this is absurd. $\square$

²This is also true for *Go* and Chess.

³Possibly finiteness condition can be relaxed.

4.2. Symmetrization. Combinatorial sequential games have a beautiful Conway recursive definition. However, for our purpose it is convenient to interpret them in terms of finite automata.

Definition 4. A combinatorial sequential game G with 2 players P_0, P_1 or Black, White, consists of:

- A deterministic automaton G , whose set of states is a set $S(G)$ called the set of game states of G .
- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \sqcup M_1(G)$: interpreted as possible moves of Black and White.
- A $\mathbb{Z}_2$ graded set of games states $S(G)$.
- Nonempty subsets $W_0(G), W_1(G), D(G) \subset S(G)$ of final (accepting) states with empty pairwise intersection, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in S(G)$, understood as the state in which the game is initialized.
- A partial game evolution function:

$$E = E_G : S(G) \times M(G) \rightarrow S(G),$$

which has the properties:

- (1) $E(s, m)$ is undefined for s in one of the accepting states.
- (2) $E(s, m)$ is defined only if m has the same grading as s .
- (3) If $E(s, m)$ is defined then it has grading $s + 1$.

We recursively define:

$$E(s, m_1, \dots, m_k) = E(E(s, m_1, \dots, m_{k-1}), m_k).$$

Definition 5. Let G be a combinatorial sequential game, $p_0 \in M_0(G)$, $p_1 \in M_1(G)$ are called **pseudo pass moves** if the following is satisfied:

- If s is not a final state and has grading $i \in \mathbb{Z}_2$, then $E_G(s, m_i)$ is defined.
- If $E_G(s, p_0, p_1, m)$ is defined then so is $E_G(s, m)$.
- If $E_G(s, p_1, p_0, m)$ is defined then so is $E_G(s, m)$.

Remark 6. Note that Go has pseudo pass moves, but chess does not, however chess can be augmented with pass moves, by also adjusting the rules so that king capture is legal and ends the game. This does not change the early-mid game dynamics in any significant way, but the end-game does change, with more tie possibilities. We will call this augmented chess.

4.3. Universal symmetrization. Let G be as above with distinguished pseudo pass moves p_0, p_1 . We say that a game history of G is **pseudo legal** if it is legal after adding or removing some pseudo pass moves to the game sequence.

Let $\mathcal{H}(G)$ be the set of finite pseudo legal game histories in G . For $a \in \mathcal{H}(G)$ and $m \in M(G)$ define $a \# m$ to be the new sequence obtained by adding the element m at the end. Define $P(a, m_0, m_1) \in \mathcal{H}(G)$ by the following prescription in decreasing priority, where the expression is defined only if the right hand side is a subset of $\mathcal{H}(G)$:

- $P(a, m_0, m_1) := a \# m_0 \# m_1 \sqcup a \# m_1 \# m_0$.
- $P(a, m_0, m_1) := a \# m_0 \# m_1 \sqcup a \# m_1$.
- $P(a, m_0, m_1) := a \# m_0 \sqcup a \# m_1 \# m_0$.
- $P(s, m_0, m_1) := a \# m_0 \sqcup a \# m_1$.

- undefined.

We then get a partial function

$$P : \mathcal{H}(G) \times M_0(G) \times M_1(G) \rightarrow \mathcal{H}(G).$$

Definition 6. Let G be a sequential game with pseudo pass moves p_0, p_1 . Its universal symmetrization (with respect to p_i) is an ssc game $\text{Sym}(G) = G'$ satisfying the following.

- (1) $B(G') = 2^{\mathcal{H}(G)}$.
- (2) $M(G') = M(G)$ as $\mathbb{Z}_2$ graded sets.
- (3) Let $\mathfrak{s} \in B(G')$, and $m_0 \in M_0(G')$, $m_1 \in M_1(G')$. Then

$$E_{G'}(\mathfrak{s}, m_0, m_1) := \cup_{a \in \mathfrak{s}} \{P(a, m_0, m_1)\},$$

(undefined if some $P(a, m_0, m_1)$ is undefined).

(4)

$$\mathfrak{s} \in W_0(G') \iff W_0(G) \cap \mathfrak{s} \neq \emptyset \wedge W_1(G) \cap \mathfrak{s} = \emptyset.$$

Similarly,

$$\mathfrak{s} \in W_1(G') \iff W_1(G) \cap \mathfrak{s} \neq \emptyset \wedge W_0(G) \cap \mathfrak{s} = \emptyset.$$

In all other cases, $\mathfrak{s} \in D(G')$.

This universal symmetrization will almost always have an impractically large number of states. This can be cut down depending on the situation, but usually this will not result in a symmetric game. So we make the following definition.

Definition 7. Let G_0 be a sequential game with pseudo pass moves p_0, p_1 . We say that G_1 is a **symmetrization** of G_0 if the following is satisfied.

- (1) G_1 is a symmetric simultaneous game.
- (2) $M(G_1) = M(G_0)$ as $\mathbb{Z}_2$ graded sets.
- (3) There is an embedding $\rho : B(G_1) \rightarrow B(\text{Sym}(G_0))$ s.t. for $s \in B(G_1)$, $E_{G_1}(s, m_0, m_1)$ is defined iff $E_{\text{Sym}(G)}(\rho(s), m_0, m_1)$ is defined, and in this case

$$\rho(E_{G_1}(s, m_0, m_1)) = E_{\text{Sym}(G)}(\mathfrak{s}, m_0, m_1),$$

for some $\emptyset \neq \mathfrak{s} \subset \rho(s)$.

Definition 8. We will say that G_1 is a **simple symmetrization** of G_1 if there is a map $\phi : B(G_1) \rightarrow B(G_0)$ s.t. for $s \in B(G_1)$, $E_{G_1}(s, m)$ is defined if and only if $E_{G_0}(\phi(s), m)$ is defined.

Let Go be the classical game with Chinese rules and standard pass moves p_0, p_1 .

Theorem 2. SGo is a simple symmetrization of Go .

Proof. The first two properties are by construction and Lemma 5. To verify Property 3 construct the map ρ as follows. Given an SGo history $a = \{m_i^0, m_i^1\}$ we get $P(\{m_i^0, m_i^1\}) \subset \mathcal{H}(Go)$ defined by recursion

$$P(\{m_i^0, m_i^1\}_{i=1}^{i=k}) = \bigsqcup_{a \in P(\{m_i^0, m_i^1\}_{i=1}^{i=k-1})} P(a, m_k^0, m_k^1).$$

$\rho(a) \subset P(\{m_i^0, m_i^1\})$ is then obtained by eliminating certain histories by way of the SGo move resolution rules, in the obvious sense.

To verify the simplicity property, define $\phi : B(G) \rightarrow B(Go)$ for example by setting $\phi(a)$ to be the Go board state obtained by replacing every stone of the

corresponding board state by a black stone. It is then immediate that Property 8 is satisfied. $\square$

Question 2. *Does augmented chess as in Remark 4 have a simple symmetrization?*

Of course augmented chess has the universal symmetrization $\text{Sym}(\text{chess})$. But it is clear, though without obvious formal proof, that Property 8 will not be satisfied. And non-locality of chess makes it difficult to reduce $\text{Sym}(\text{chess})$ to something that does satisfy Property 8.

REFERENCES

- [1] E. R. BERLEKAMP AND D. WOLFE, *Mathematical Go: Chilling Gets the Last Point*, AK Peters, Natick, MA, 1994.
- [2] M. A. HUGGAN, *Studies in Alternating and Simultaneous Combinatorial Game Theory*, ph.d. thesis, Dalhousie University, Halifax, Nova Scotia, Canada, 2019.
- [3] J. F. J. NASH, *Equilibrium points in n -person games*, Proc. Natl. Acad. Sci. USA, 36 (1950), pp. 48–49.
- [4] R. PENROSE, *Emperor’s new mind*, 1989.
- [5] J. M. ROBSON, *The complexity of go*, Proceedings of the International Congress of Mathematicians, (1983), pp. 661–666.
- [6] U. SCHWALBE AND P. WALKER, *Zermelo and the early history of game theory*, <http://www.math.harvard.edu/~elkies/FS23j.03/zermelo.pdf>.